Yusuf Afifi

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Education

University of California, Berkeley — Berkeley, CA
Master of Engineering in Electrical Engineering and Computer Science — *2026 – 2027*

- Coursework (Fall 2026): CS 289A Machine Learning, CS 280 Computer Vision.

University College London — London, UK
*BSc (Hons) Computer Science, Minor in Applied Mathematics — **First Class Honours*** — *2023 – 2026*

- Dissertation (supervised by Yun Fu): **IntelliTrade**, multi-agent LLM signals over a **Neo4j** causal-temporal knowledge graph feeding a dual-stream **DQN**; **64.6%** mean cumulative return vs. 45.5% for the S&P 500 (walk-forward, Jan 2024 – Feb 2026), beating it on **8 of 9** tickers; LLM signals added **22.4 points** over a price-only ablation.
- TA for COMP0002 (C), COMP0004 (Java), ENGF0034 (Python), and COMP0016 (Systems); ∼50 students each.

Experience

AI Research Fellow | *RL, Post-training* — San Francisco, CA
Principle — *Jun 2026 – Sep 2026*

- **First author**, *Do Your Own Research: Learning to Forecast by Learning to Search*, *NeurIPS 2026 Workshop on Foundation Models for Temporal Systems* (code): agentic RLVR that puts leak-filtered web search inside the RL loop, so the model learns to research before it forecasts.
- Single-epoch LoRA **GRPO** on Qwen3.5-35B-A3B cut calibration error (ECE) **31%** and searches per rollout from 3.8 to 2.25; outperformed four frontier models on soft Brier, including Claude Opus 4.5 (**0.254** vs. 0.256), at ∼**5%** of its per-question cost.
- Built the RL environment (Environments Hub) from **2,100+** resolved Polymarket questions with Prime Intellect's **verifiers**, plus the search harness; prototyped **Qwen3.5-9B** on **vLLM** on one **48GB L40S**, then trained at **35B** with **prime-rl**.

Software Development Engineer Intern | *LLM Agents* — London, UK
Amazon Prime Video — *Jun 2025 – Sep 2025*

- Built a text-to-SQL agent (**Claude** on **Bedrock**, **Kendra** index of schemas and business logic, **Redshift**) that cut ad-hoc reporting time **75%** for SVOD accounting, handling multi-table joins with FX normalization; validated at >**90%** accuracy over **300+** accountant-vetted questions.
- Shipped it as an internal web app used by **200+ weekly active users** across global accounting teams, with a **React** frontend over a **Lambda**-proxied WebSocket API and a human-in-the-loop approval step.
- Demoed Amazon-wide at Prime Video's Spark AI Demo Day; deployed with **AWS CDK** and CI/CD under least-privilege IAM, with **CloudWatch** alarms on token spend and latency.

Projects & Research

Agency | *Sky Computing Lab, advised by Prof. Alvin Cheung* — Aug 2026 – Present

- Built self-context-editing tools for the lab's open-source agent profiling platform, letting agents list, read, rewrite, and revert blocks of their own context window across native, **Claude Code**, and **Codex** harnesses.
- On the long-context **Oolong** benchmark (RLM-style tasks), cut cost per run **54%** by replacing compaction's full cache wipes with edits that invalidate only the changed blocks.
- Unified token accounting across OpenAI, Anthropic, and Bedrock so all **6 harnesses** compact at ∼**90%** of the real context limit (previously never, or at 40–65%); validated on the full **1,700+**-test suite and live EC2 runs.
- Benchmarked Agency on **C-to-Rust** translation of **SQLite** for **DARPA TRACTOR**, profiling a matrix of harnesses and models to find the most efficient configuration; cut token usage **30%** vs. earlier translations.

ReadingStar: On-Device Speech AI with Intel | *Whisper, ONNX, OpenVINO* — Apr 2025

- Owned the speech-model side of an offline karaoke app built with **Intel** and the **National Autistic Society** for speech development in neurodiverse students: the first education app optimized for NPUs, distributed to UK schools (e.g. Helen Allison) and featured by Intel.
- Ran **Whisper** in **real time**, fully offline, for live lyric alignment and scoring: exported it to **ONNX**, quantized it to **INT8** with **OpenVINO**, and moved inference onto the **Intel NPU**, benchmarked at **88% lower energy** per inference than CPU.